

PROJECT SUMMARY (POINT-WISE)

Customer Purchase Propensity Analysis Project

Task 1: Project Planning & Problem Framing

- Defined business objective: predict customer purchase behavior.
 - Framed problem as a **binary classification task (0/1)**.
 - Identified key drivers: age, income, spending, and transaction patterns.
 - Selected relevant ML approach for predictive modeling.
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Task 2: Data Import & Understanding

- Loaded datasets from **CSV, JSON, and SQL sources**.
 - Integrated data using common keys (customer_id, product_id).
 - Performed initial inspection using .info(), .head(), and .describe().
 - Verified data structure, types, and missing values.
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Task 3: Exploratory Data Analysis (EDA)

- Conducted **univariate analysis** to understand feature distributions.
- Performed **bivariate analysis** to study relationship with target variable.
- Used **multivariate analysis** (heatmaps, pairplots, correlations).
- Identified key patterns in customer income, spending, and purchase behavior.

Task 4: Missing Data Handling

- Applied multiple imputation techniques:
 - Simple Imputer (mean & mode)
 - KNN Imputer
 - MICE (Iterative Imputation)
 - Random Sample Imputation
 - Used Missing Indicator to capture missingness patterns.
 - Compared with Complete Case Analysis for baseline evaluation.
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Task 5: Outlier Detection & Handling

- Detected outliers using:
 - Z-score method
 - IQR method
 - Percentile method
 - Applied Winsorization to cap extreme values.
 - Reduced impact of skewed and extreme observations.
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Task 6: Mixed & Date-Time Variable Handling

- Converted date columns into datetime format.
 - Created new feature: **days_since_signup**.
 - Engineered customer tenure-based insights.
 - Cleaned inconsistent or mixed-format customer IDs.
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Task 7: Encoding Categorical Data

- Applied **Label Encoding** for binary/low-cardinality variables.
- Used **One-Hot Encoding** for nominal categorical features.

- Applied **Ordinal Encoding** for structured variables (education, satisfaction).
 - Performed **binning on income** for segmentation.
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Task 8: Feature Scaling

- Applied multiple scaling techniques:
 - StandardScaler
 - MinMaxScaler
 - MaxAbsScaler
 - RobustScaler
 - Normalizer
 - Used ColumnTransformer for parallel transformations.
 - Compared scaling approaches for model readiness.
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Task 9: Feature Engineering & Transformation

- Created interaction features (purchase_per_day).
 - Applied transformations:
 - Log transformation
 - Square root transformation
 - Reciprocal transformation
 - Power transformation (Yeo-Johnson)
 - Performed binning (income groups).
 - Created binary feature (frequent_buyer).
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Task 10: Final Output & Export

- Exported final dataset as **processed_customer_data.csv**.
- Saved preprocessing pipeline for reuse.
- Documented full workflow and transformation steps.
- Prepared dataset for machine learning modeling stage.

FINAL OUTCOME

- Cleaned dataset
- Feature-engineered dataset
- Fully ML-ready structured pipeline
- End-to-end data science workflow completed